

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 13

Organize and Layout Windows and Pages

Step 14

Testing



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THE USER INTERFACE (UI) DESIGN PROCESS

Step 13

Organize and Layout Windows and Pages



Tujuan

- Setelah mengikuti materi ini mahasiswa dapat:

Mahasiswa/i dapat mampu menerapkan pengaturan dan tataletak Windows dan halaman web yang sesuai dan bermakna.

Objectives of Windows/Pages Layout and Organization

Opening

Pengaturan/
Pengorganisasian Layout
windows dan halaman
web yang baik



**Mempercepat Pemahaman
Informasi**

**Mempercepat Proses
Eksekusi Task dan fungsi**

**Meningkatkan User
Acceptance**

Our Goal

- Organizing and laying out graphical and web screens to encourage quick and accurate information comprehension and control execution
- Organizing Meaningfully and efficiently
- Creating Grouping
- Providing Allignment and balance

Guidelines for Deciding Pages Layout

General Guidelines

(Review step 3)

Amount of information

- Too much is confusing, too little is inefficient
- Proportion of information to more than 30-40% of the window entire area
- Leave sufficient margin around all screen elements and between elements and the screen border

Flow

- Provide ordering that logical, sequential, and rhythmic
- Maintain a top down, left to right flow

Navigation

- Efficiency, minimize cursor and pointer travel, minimize hands travel of user from mouse to keyboard
- Count the tabs, count the "click"

Visually pleasing composition

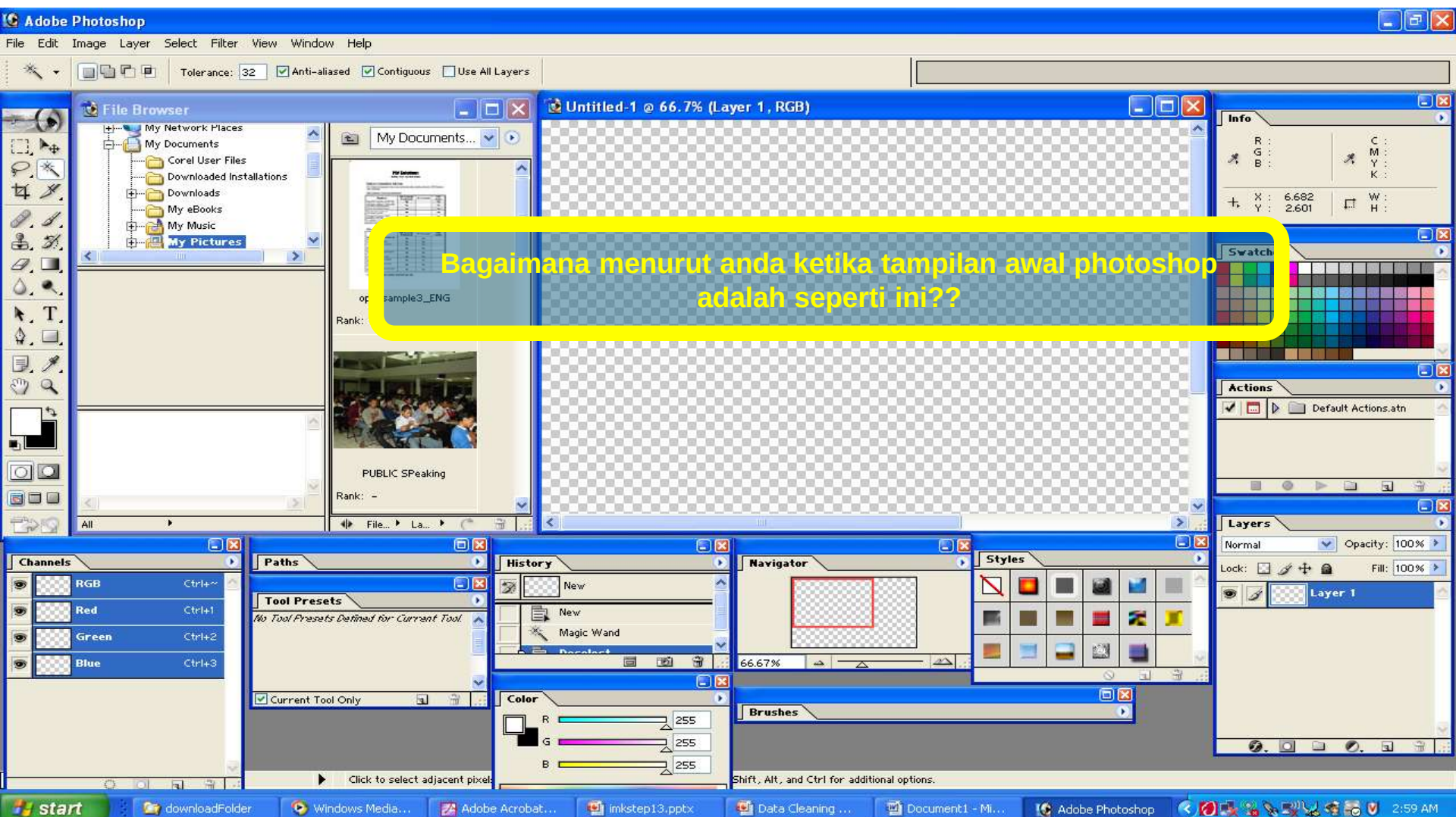
- Balance, grouping and alignment
- Adequate use of white space

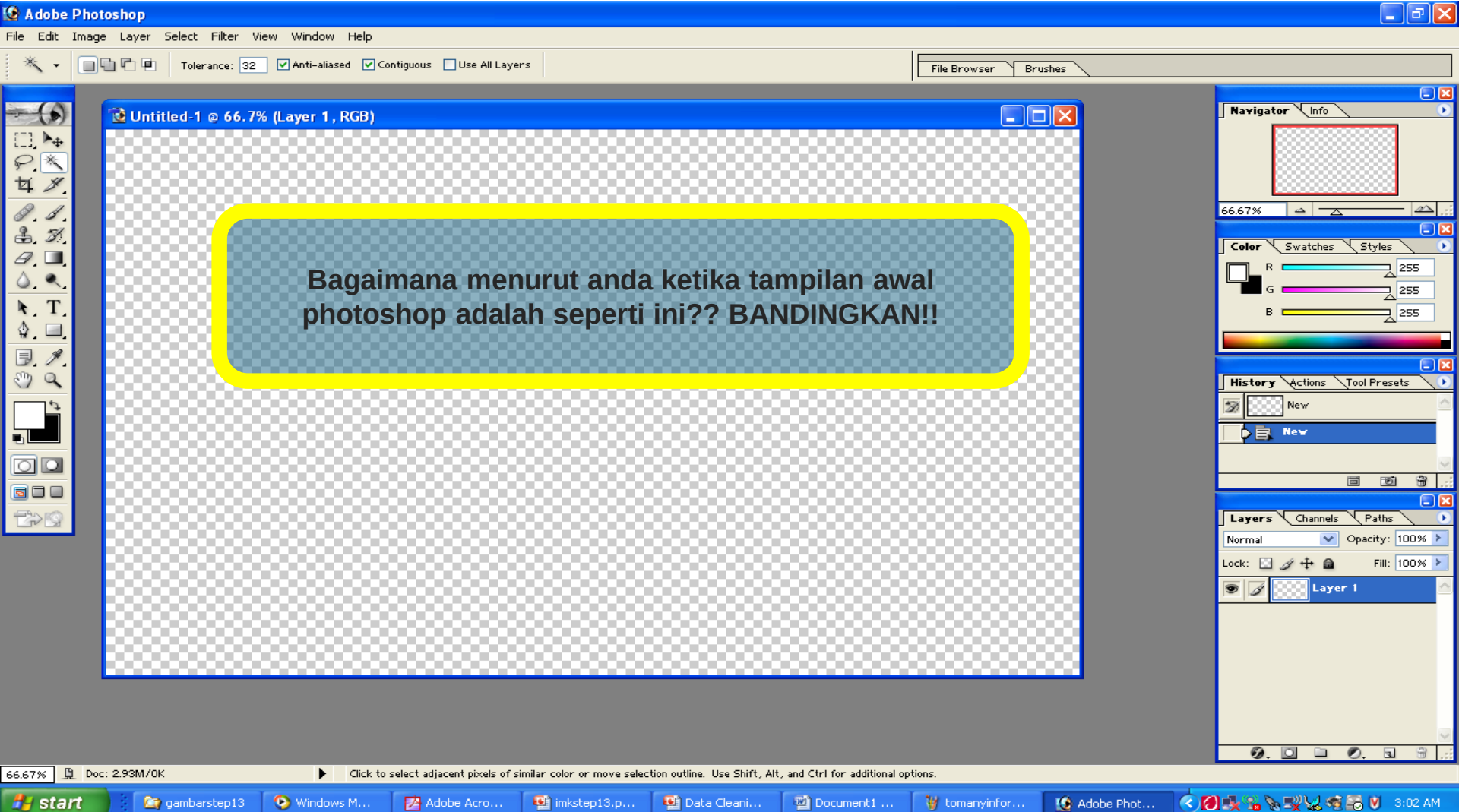
Consistency

- Provide consistency with user experiences and cultural convention
- Provide consistency on operational and navigational procedure
- Consistency on presentation

Proper Emphasis of element

- Critical, important, secondary, peripheral





Organization Guideline (Creating Grouping)

General

- Provide grouping of associated elements

White Space

- Provide adequate separation of grouping through the use of white space
- Leave adequate space around groups or related control, between grouping and windows border
- The space between grouping should be greater than the space between fields within a grouping

Headings

- Provide section heading and subsection heading for multiple control grouping
- Provide heading that meaningfully and concisely describe the nature of the group of related field

Borders

- Enhance grouping through incorporation of borders around element of single control, groups of a related control or field
- Individual control border should be visually differentiable from border delineating groupings of field or control
- Do not place border around command button
- Do not place individual field or control border around single entry field, single list box, single combination boxes, single spin boxes, single sliders

Organization Guideline (Creating Grouping)



Municipality: ☐ City
☐ Township
☐ County
☐ State

Building:

Floor:

Telephone:

Figure 13.1: Groupings.

Municipality: ☐ City
☐ Township
☐ County
☐ State

Building:

Floor:

Telephone:

Department: ☐ Administration
☐ Finance
☐ Fire
☐ Police
☐ Public Works
☐ Social Services

Manager:

Employees:

Payroll:

Figure 13.2: Groupings using white space.

Organization Guideline (Creating Grouping)

JURISDICTION

Municipality: ☐ City
☐ Township
☐ County
☐ State

LOCATION

Building:
Floor:
Telephone:

PERSONNEL

Department: ☐ Administration
☐ Finance
☐ Fire
☐ Police
☐ Public Works
☐ Social Services

Manager:
Employees:
Payroll:

Figure 13.3: Groupings with section headings.

Organization Guideline (Creating Grouping)



Contents

☒ Preface
☐ Illustrations
☐ Index
☒ Bibliography

Contents:

☒ Preface
☐ Illustrations
☐ Index
☒ Bibliography

Figure 13.4

— If the control caption exceeds the length of the choice descriptions, extend the border two character positions to the right of the caption.

Justification

☐ None
☒ Left
☐ Center
☐ Right

Figure 13.5

Department:

- ☐ Administration
- ☐ Finance
- ☐ Fire
- ☐ Police
- ☐ Public Works
- ☐ Social Services

Department

- ☐ Administration
- ☐ Finance
- ☐ Fire
- ☐ Police
- ☐ Public Works
- ☐ Social Services

- Department:
- ☐ Administration
 - ☐ Finance
 - ☐ Fire
 - ☐ Police
 - ☐ Public Works
 - ☐ Social Services

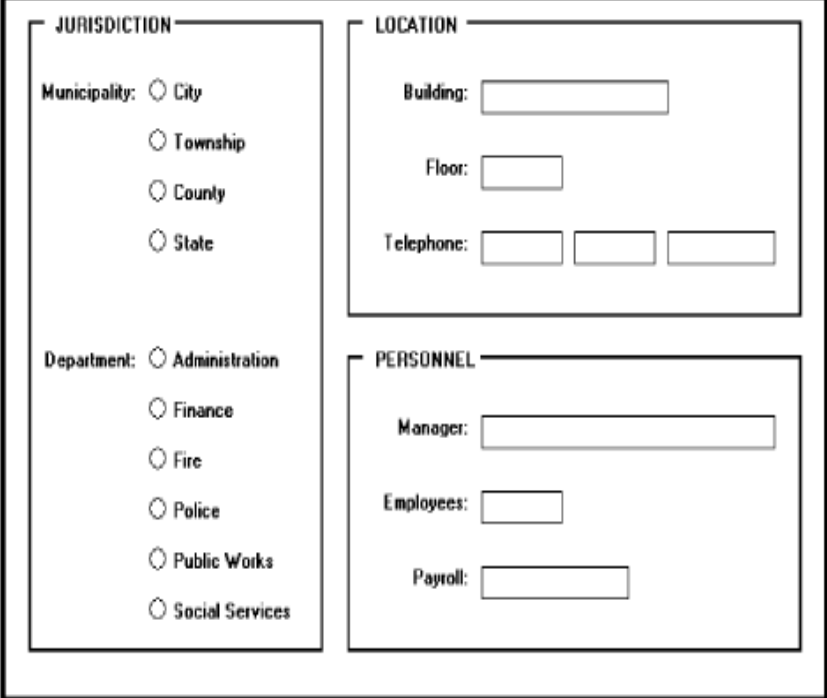
- Department:
- ☐ Administration
 - ☐ Finance
 - ☐ Fire
 - ☐ Police
 - ☐ Public Works
 - ☐ Social Services

Figure 13.6: Examples of vertically arrayed controls without and with borders.

Organization Guideline (Creating Grouping)

Section Borders

- Incorporate a thicker single-line border around groups of related entry or selection controls.
- For spacing,
 - Vertically, leave one line space between the top and bottom row of the entry or selection control elements.
 - Horizontally, leave at least four character positions to the left and right of the longest caption and/or entry field.
- Locate the section heading in the top border, indented two character positions from the left border.



JURISDICTION

Municipality: ☐ City
☐ Township
☐ County
☐ State

Department: ☐ Administration
☐ Finance
☐ Fire
☐ Police
☐ Public Works
☐ Social Services

LOCATION

Building:

Floor:

Telephone:

PERSONNEL

Manager:

Employees:

Payroll:

Figure 13.8: Grouping of sections using borders.

Organization Guideline (Creating Grouping)

DOCUMENT

Justification:

☐ None
☐ Left
☐ Center
☐ Right

Contents:

☐ Preface
☐ Illustrations
☐ Index
☐ Bibliography

Title:

Number of Chapters:

Number of Pages:

Figure 13.9: Differentiable control and section borders.

Text Box:

Combo Box: 

Attached Combo:

List Box:

Spin Box:

Button Button Button

Figure 13.10: Kinds of borders to avoid using.

Web Page Guidelines

General

- Efficient
- Logical and consistent
- Self explanatory
- Scannable
- Proper Balance

Layout Grid

- Establish a standard layout grid
- Consistency in every page
- Better alignment and visual connection between element

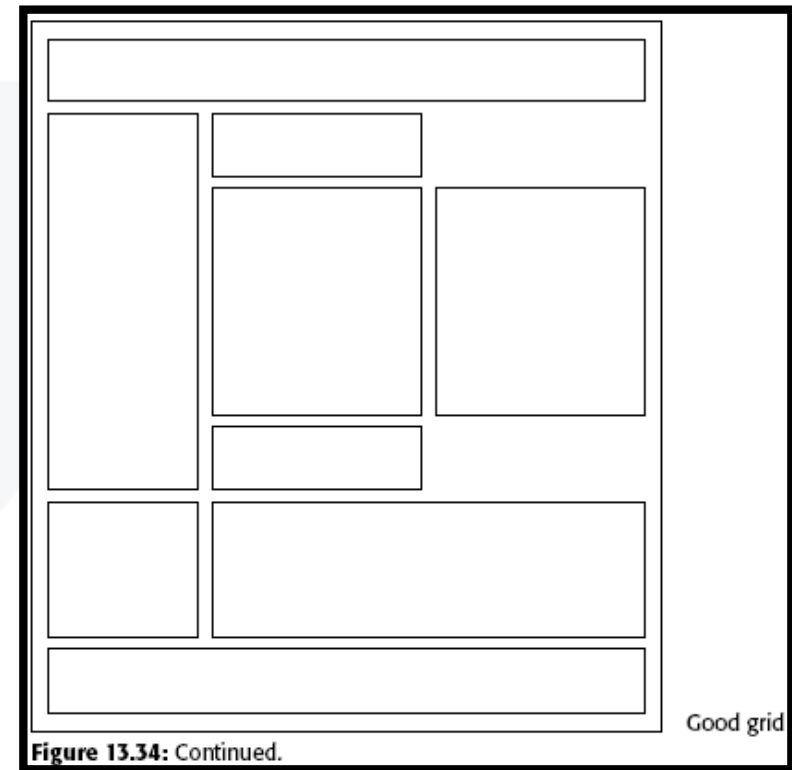
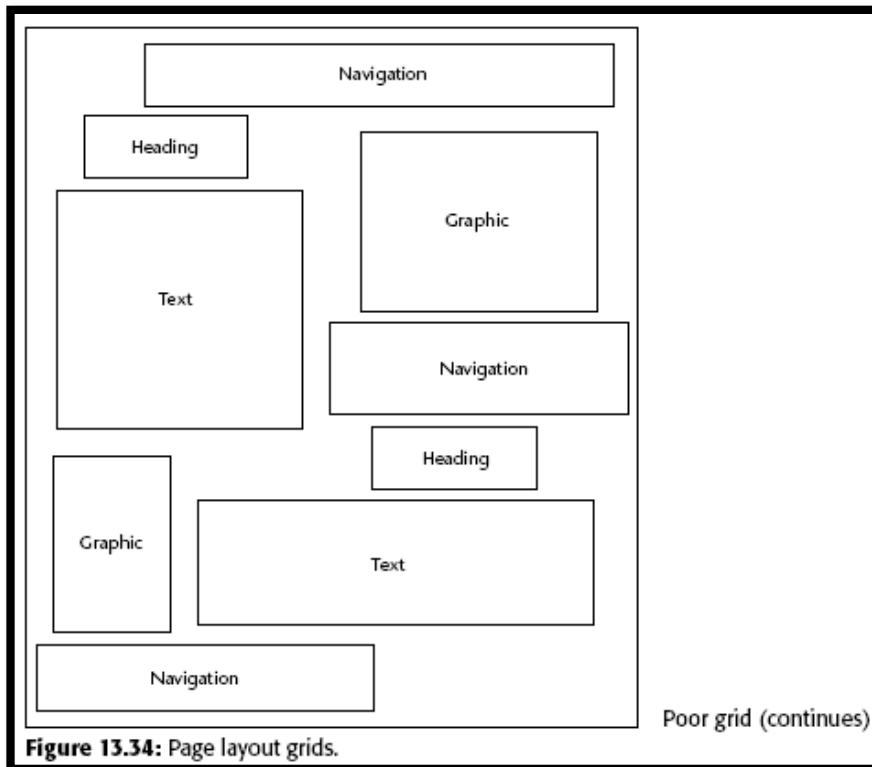
Element

- To keep pages simple, restrict the number of distinct functional element on a page, element include such tittle, navigation bar, textual link to major site area, textual content, graphics and page footer

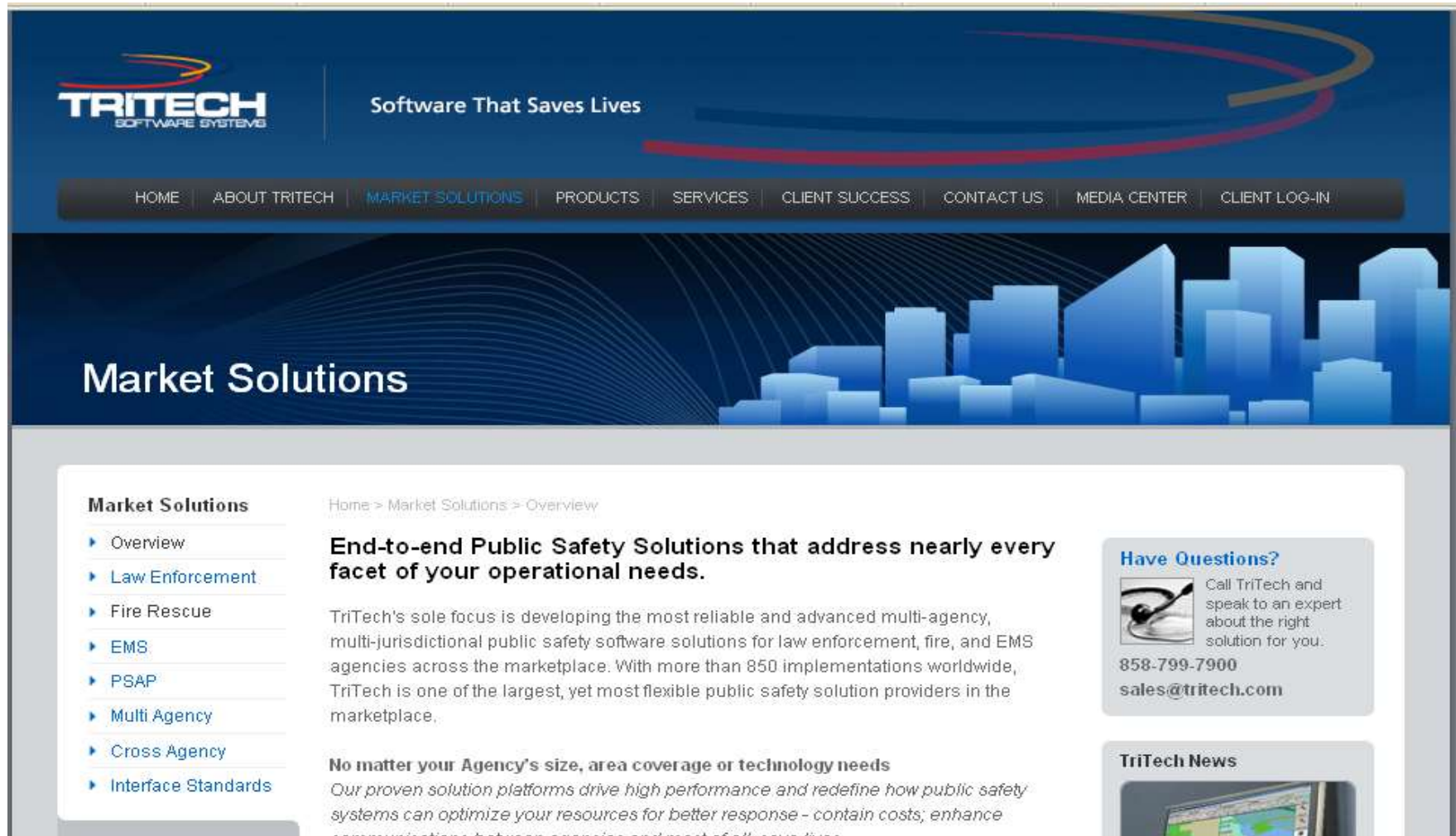
Size

- Page Length: Minimize page length, restricting it to two or three screens of information
- Anticipate Page Breaks
- Horizontal scrolling: Layout page so horizontal scrolling is never necessary

Grid Example (Poor VS Good)



What do you think about
this GRID??



The screenshot shows the TriTech Software Systems website. The header features the TriTech logo and the tagline "Software That Saves Lives". A navigation bar includes links for HOME, ABOUT TRITECH, MARKET SOLUTIONS, PRODUCTS, SERVICES, CLIENT SUCCESS, CONTACT US, MEDIA CENTER, and CLIENT LOG-IN. The main content area is titled "Market Solutions" and features a sidebar with a list of solution categories: Overview, Law Enforcement, Fire Rescue, EMS, PSAP, Multi Agency, Cross Agency, and Interface Standards. The main text area highlights "End-to-end Public Safety Solutions that address nearly every facet of your operational needs." and describes TriTech's focus on developing multi-agency software solutions. A "Have Questions?" section provides contact information: 858-799-7900 and sales@tritech.com. A "TriTech News" section is also visible at the bottom right.

TRITECH
SOFTWARE SYSTEMS

Software That Saves Lives

HOME | ABOUT TRITECH | [MARKET SOLUTIONS](#) | PRODUCTS | SERVICES | CLIENT SUCCESS | CONTACT US | MEDIA CENTER | CLIENT LOG-IN

Market Solutions

Market Solutions

- ▶ Overview
- ▶ Law Enforcement
- ▶ Fire Rescue
- ▶ EMS
- ▶ PSAP
- ▶ Multi Agency
- ▶ Cross Agency
- ▶ Interface Standards

Home > Market Solutions > Overview

End-to-end Public Safety Solutions that address nearly every facet of your operational needs.

TriTech's sole focus is developing the most reliable and advanced multi-agency, multi-jurisdictional public safety software solutions for law enforcement, fire, and EMS agencies across the marketplace. With more than 850 implementations worldwide, TriTech is one of the largest, yet most flexible public safety solution providers in the marketplace.

No matter your Agency's size, area coverage or technology needs

Our proven solution platforms drive high performance and redefine how public safety systems can optimize your resources for better response - contain costs; enhance communications between agencies and most of all, save lives.

Have Questions?

 Call TriTech and speak to an expert about the right solution for you.

858-799-7900
sales@tritech.com

TriTech News



What do you think about
this GRID??



Web Page Guidelines (cont..)

Organization

- Important element at the top
- Positioning
- Reduced Complexity

Formatting

- White Space
- Textual Line Length
- Text
- Better alignment and visual connection between element

Multiple Audience

- Present information in different level of detail

Platform

- Anticipate the typical screen and resolution and specify an image save area of the screen that will always be visible to most viewer

Web Page Guidelines (cont..)

Navigation element

- Provide a global navigation bar at the top of each page
- Provide a local category or topical links navigation bar on the left side of a page
- Optionally, provide a secondary navigation column on the right side of the page
- Place minor illustrative, parenthetical, or footnote links at the end
- For long pages provide “list of contents” link
- Create a common and consistent theme
- Never create pages without navigational options

Homepage

- Limit to one screen
- Clearly identify the web site content and purpose
- Element to include: Name or logo and tagline, web site name, brief desc of web site, Summary of the key informational content, Navigational links to most of the site of major section
- Site map
- Summary of the latest news or promotion
- Search facility

Layout Examples

 **Sponsors of Tomorrow.™**

Sensory robotics. One of many incredible projects going on inside Intel.

SHARE >



THE FUTURE OF TRANSPORTATION

Use your imagination to design a futuristic dream car.

▶ CREATE YOUR CAR



Enter The Time Machine > | Share Your Vision > | Meet The Rock Stars > | Get Computer Smart > | Find A Processor >



Sponsors of Tomorrow.™

Sensory robotics. One of many incredible projects going on inside Intel.

SHARE >



THE FUTURE OF ENVIRONMENT

Get a glimpse of a future when audio and video sensors will "see" our world better than we do.

▶ WATCH VIDEO

Lily Mummert
Research Scientist



Enter The Time Machine >

Share Your Vision >

Meet The Rock Stars >

Get Computer Smart >

Find A Processor >

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 Games Videos Create My Page Characters Movies TV Music Live Events Parks Store For You

All Games Arcade Puzzle Word Action Sports Characters Video Games



Games

Today in Games

Master the element of *fire*



Top Five Games

- [1. Hannah Montana Rock the Beat](#)
- [2. Club Penguin](#)
- [3. Scrooge's Soarin' Star Ride](#)
- [4. Haunted Mansion: Black Widow Bride](#)
- [5. Old Dogs Golf Cart Chaos](#)

Virtual Worlds





Scroll down!!

Interaksi Manusia dan Komputer (IMK)

THE USER INTERFACE (UI) DESIGN PROCESS

Step 14
Testing



Tujuan

- Setelah mengikuti materi ini mahasiswa dapat:
Mahasiswa/i dapat memahami konsep pengujian dalam desain UI dan cara mengukur UI

Usability Testing



Evaluation Of Usability

- There are different approaches to evaluating usability, three main categories (Chandler&Hyatt,2002) :
 - **Inspection methods** : do not involves customer. Heuristic evaluations and walk-throughs
 - **Lab testing methods** : involves the testing of representatives users.
 - **Surveys and customer reporting methods** : provides subjective data and is based on expressed feelings, attitudes, and perceptions of the interaction's usability and overall desirable qualities.

Types of Usability Testing

Heuristic Evaluation

- Based on the expertise of a set reviewers.
- Generally apply across situations and are used to predict potential problems with the interaction.
- Has the advantage of being easy to use and does not require expensive lab settings.
- Kalbach (2007) identifies 3 steps for heuristic evaluation :
 - 1.Prepare : Agree on who will do the review.
 - 2.Execute : Go through the interaction prototype, focusing on one principle at a time.
 - 3.Consolidate : Discuss your findings with other reviewers.

Heuristic Evaluation (Cont)

- Nielsen (1994) proposed 10 general “Heuristics”
 - :
 - Visibility of system status
 - Match between system and the real world
 - User control and freedom
 - Consistency and standards
 - Error prevention
 - Recognition rather than recall
 - Flexibility and efficiency of use
 - Aesthetic and minimalist design
 - Help users recognize, diagnose, and recover from errors
 - Help and documentation

Heuristic Evaluation (Cont)

- Kalbach(2007) proposed the following heuristic specific to web based interactions :
 - Balance
 - Ease of learning
 - Efficiency
 - Consistency
 - Clear labels
 - Orientation
 - Exploration
 - Differentiation
 - Information use
 - Modes of searching

Heuristic Evaluation (Cont)

- Heuristic evaluation can be also accelerated with the use of checklists.
- A sample checklist can be found at <http://www.stcsig.org/usability/topics/articles/he-checklist.html>.
- Given the high cost of conducting interaction evaluations, new trend is towards automated evaluation. For example, free tools include :
 - Wave : <http://wave.webaim.org>
 - InFocus : www.ssbtechnologies.com/products/infocus

Walk-throughs

- Walk-throughs simulate a customer's experience with the interaction.
- The results of the simulation are compared with the goals, expectations, and knowledge that the first-time customer is expected to have with an online store

Web Accessibility

- It is about building web sites, applications, and pages that present few barriers to use are possible for anyone, regardless of ability and the device used to access the information.
- Web accessibility should be :
 - Accommodate persons with disabilities (visual disabilities, hearing disabilities, Physical disabilities and Cognitive and neurological disabilities)
 - Provide access to those using slower connections that normally have the images tuned off as well as increase interoperability with mobile devices.

Web Content Accessibility Guidelines (WCAG)

- **WCAG 1.0**
 - Guidelines 10 : Use interim solutions.
 - Guidelines 11 : Use W3C technologies and guidelines.
 - Guidelines 12 : provide context and orientation information.
 - Guidelines 13 : provide clear navigation mechanism.
 - Guidelines 14 : ensure documents are clear and simple.
- **WCAG 2.0**
 - Content must be perceivable.
 - User interface components in the content must be operable.
 - Content and controls must be understandable.
 - Content must be robust enough to work with current and future technologies.
- See [www.w3.org/WAI/GL/2005/06/30-mapping .html](http://www.w3.org/WAI/GL/2005/06/30-mapping.html)

Usability Testing

- Answer four key questions about an interaction.
 1. Learnability - how easy does the learning take place?
 2. Throughput - how easy is it to use ?
 3. Flexibility - how easy is it to change in environment and tasks?
 4. Attitude – doe it provide the user with a positive attitude?

Usability independent and dependent variables (Eason, 1988)

Independent Variables	Dependent variables
User Characteristic : Knowledge Discretion Motivation	User reaction-cost benefit analysis (positive outcome) Good task system match Continued user learning
System function: Task match Ease of use Ease of learn	User reaction-cost benefit analysis (negative outcome) Restricted use Non-use, partial use Distance use
Task characteristic : Frequency Openness	

Usability Measures (Shackel, 1990)

Measure	Comments
Effectiveness	At better than some required level of performance(in terms of speed and errors) By some required percentage of the specified target range of users Within some required proportion of the range of usage environments
Learnability	Specified time from installation and start of training Based upon some specified amount of training Some specified relearning time for intermittent users
Flexibility	Allowing adaptation to some percentage variation in task beyond those first specified(difficult to determine in practice and acan be replaced by usefulness-achievement of user's goals)
Attitude	Within acceptable levels of human cost in terms of tiredness, discomfort, frustration, and personal effort

Steps of Testing

Conducting the usability test

- First conducting is a pilot test.
- A pilot test allows to “debug” the test and find out if there any initial problems with the interaction under test.
- Usability tests might be conducted in usability testing laboratories. If labs are not available another viable solution is go to the customer’s location.
- Users can be interviewed as they are performing the task. Make sure the questions are specific so the tester can get the answers.

Analyzing and Presenting Usability Test Results

- Data collected from usability test (Dumas&Redish,1999)
 - A list of problems from the test
 - Quantitative data on time, errors, and other performance measures, including subjective ratings on questionnaires
 - Tester's comments from logs, notes, and questionnaires
 - The testing team's written notes
 - Background data on the participants
 - Videotapes of the test, perhaps from several different viewpoints in the room
- That the analysis should follow some guidelines to make statistical analysis as relevant as possible :
 1. Use inferential statistics only if you understand how to apply and interpret them
 2. After you employ a statistical test, carefully explain what the test means
 3. Describe your interpretation of key data values when you don't compute statistical tests.

Analyzing and Presenting Usability Test Results (Cont)

- Both your quantitative data analysis as well as the qualitative data from feedback and notes will help you organize the information into two areas :
 - Scope – How widespread is the problem?
 - Severity – How critical is the problem ?
- A final report should be prepared to summarize the results of the test. The following sections for a usability test :
 - Procedures
 - Evaluator profiles
 - Observations
 - Evaluator quotes
 - Conclusions
 - recommendations
- A sample usability report and data analysis can be downloaded from the link below :
 - <http://www.utexas.edu/learn/usability/report.html>